

The word animation is extracted from a Latin language, "Anima" which means "Soul". The animation is a technology that allows many images to create animation. The images which have a different location, put it together one after another at a high speed the images look as if they are moving. Any image object an entity to walk, talk, or do some movement that cannot be physically moved.

Q. Define Animation.

- o Animation is the process of making something alive. it is an art to produce moving images. when we use animation in a movie or film, it is called computer generated image.
- o it is the process of making static image that looks like moving image.
- o it is just a continuous series of still images that are displayed in a sequence.
- o it is concerned with the visual or aesthetic aspect of the project.
- o Animation is an object moving across or into or out of the screen.

Q. What are principles of animation?

- o Animation is possible because of:
 - o Biological Phenomenon
 - Persistence of vision - it is a type of optical illusion
 - o Psychological Phenomenon
 - it leads the brain to see a moving image. called Phi - object from a series of images
- ⇒ An object seen by the human eye remains chemically mapped on the eyes retina for a brief time after viewing.
- ⇒ it is a series of images that changed sequentially to give illusion of moving images.

Animation Space

2D - Space - it is flat animation

⇒ In 2D space, the visual changes being an image alone occurs on flat

Coordinate x and y axis. It uses vector
as animation goes to produce and edit animated images.
e.g. blinking word, color cycling logo,
cel animation, cartoons (Dorothy Dukes)

Software:

Flash, PowerPoint, Adobe Photoshop & Sketch

⇒ 2D Animations are very simple and static

2.5D animation: flat 2D animation with 3D effects.
Easy to control - less cost
Less time consuming - easy to change

⇒ 2.5D animation is a 2D animation drawn into a 3D space as it involves the motion

of 2D-animated object in a 3D space.

⇒ Animators use the impression of drawing 2D objects in motion that appears to be moving in a 3D space accomplished by using brilliant art skills of layering, shadows, perspective animations, morphing and several other techniques.

⇒ 2.5 Animations feels like it has some kind of depth.

⇒ An illusion of depth is created through shadowing, highlighting, and forced

perspective. Though in reality the images exist in 2D dimensions.

⇒ Also known as Pseudo 3D - animation.

3D - Space

⇒ Complicated and realistic animations are done in 3D space.

⇒ 3D animation gives us a feeling that there is 100% depth from every angle that camera follows.

⇒ It allows us to create realistic objects.

⇒ Textures and lighting are applied to the objects.

⇒ Lot of effort, resources and animation time greatly needed towards building a successful 3-D animation.

What are steps that should be followed in creating animation?

⇒ The following steps need to be followed:

- o Organize the creation in a series of logical steps.
- o Choose an animation tool best suited for the job.

- o Build and tweak the sequences.
- o Post-process the completed animation.

The process of producing so animation is achieved into the path modeling layout & animation created

Write the name of animation techniques

- ⇒ Cell Animation & Path Animation
- ⇒ Computer Animation
- ⇒ Animation Process

What is cell animation?

⇒ The term "cell" is derived from the clear celluloid sheets that were used for drawing each frame.

⇒ It is traditional form of animation.

⇒ Cell animation is the art of creating plastic called "cells".

⇒ Animators transfer draft drawings onto transparent sheets of plastic called cells.

⇒ Each cell features one drawing with an outline on one side of the plastic and the color is filled in on the other.

⇒ These 'cells' are then placed over a background and photographed in sequence.

⇒ When played at a speed of 12 or 24 frames they create the illusion of movement.

⇒ Today's cell animation begins with

Define keyframes.

Keyframes

Keyframes refer to the first and the last frames of an action in animation.

It defines the starting and ending point of any smooth transition.

A key frame is a point at which we can change an effect or animation value.

For example: If we want to change the color of an object from blue to red.

This can be done by changing the value of one or more keyframes in competition timeline.

- Most imp part in animation

⇒ Define character poses at specific time steps called keyframes.

Define Tweening.

Tweening Also known as "inbetweening".

⇒ It is the process of generating keyframes that go



⇒ Tweening is necessary to convey a sense of fluid movement with still images.

⇒ Tweeners are typically considered less repetitive than keyframes

⇒ it is typically made by junior artists

⇒ it is the process of creating intermediate frames called inbetweens between two keyframes. It fills the gap between the two keyframes.

⇒ It gives the impression that the first image has smoothly evolved into the second one.

⇒ Flash can create two types of tweened animation:
 o Motion tweening
 o Shape tweening

Shape tweening is useful when you want to transform one shape into another. It is useful when you want to transform one shape into another.

Q. What is pencil test?

⇒ Pencil test is an early version of animated scene.

⇒ A pencil test is the process where an animator sketches out the scenes and the characters, to check how the animation is working.

⇒ It helps animators see if they have timed their animation correctly and how a character's acting looks before committing to the next steps of clean-up and coloring in.

A. Define kinematics and inverse kinematics.

Kinematics: These are mathematical methods used to describe the motion and position of an object in a virtual environment.

⇒ Kinematics is the study of motion and position of an object in a virtual environment. It is the study of motion and position of an object in a virtual environment.

Inverse kinematics:

⇒ Inverse kinematics is the process of linking objects and defining their relationship and limits.

- used to control the movement of objects

Define mapping

Mapping is the process of mapping one set of data to another. It is the process of mapping one set of data to another.

⇒ it transform object shape from one form to another.
 o after used to change from one face to the other face.

⇒ Morphing is short transform one image into another.

⇒ Morphing is an effect that can be used to manipulate still images or to create interesting animated transformations.

⇒ Morphing allows you to smoothly blend two images so that one image seems to melt into the next often producing some amusing results.

Morphing vs Warping

<u>Morphing</u>	<u>Warping</u>
• Transforming one image to another through a series of intermediate images.	• Warping is the distortion of one static image to produce another static image.

Q What are file formats used in animation?

- .dis and .dcs (Director files)
- .fli and .flc (Animator Pro files)
- .man (3D Studio Max files)
- .pics (SuperCard and Director files)
- .fla and .swf (Flash files)
- .svg

Discuss the physical & psychological principles as to why computer animation works as well as how it is usually presented. Briefly discuss the origin of cell animation & Concept go in it?

Def:

- Animation -

Animation is the art of making something come alive.

Animation is a continuous series of still images that are displayed on a screen in a sequence.

it is the process of making static image as alive.

it is a visual change over time and can add great power to your multimedia projects and web pages.

Depending on the size of the project, you can animate the whole thing or can just animate parts of it.

- Visual effects such as wipes, Dissolves and dissolves, available in most editing packages are a simple form of animation.
- Animation is an object actually moving across, into, or out of the screen.

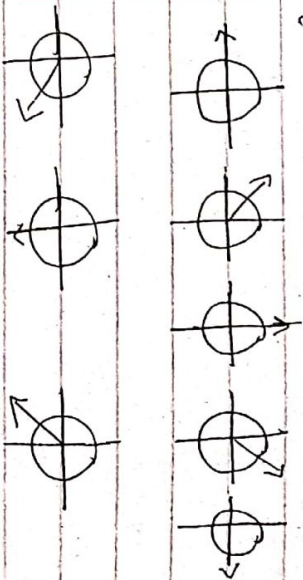
Principles of Animation

- ⇒ Animation is possible because of:
 - Biological phenomenon known as persistence of vision.
 - A psychological phenomenon known as phi.
- ⇒ An object seen by human eye remains chemically mapped on the eye's retina for a brief time after viewing.
- ⇒ Combined with the human eye need to conceptually complete a perceived action.
- ⇒ With animation, a series of images are changed very slightly and very rapidly, one after the other,

Semingly blending together into a visual illusion of movement.

⇒ Digitally display video builds at 24, 30, or 60 entire frames or pictures every second.

⇒ Movies on film are typically shot at a shutter rate of 24 frames per second.



Animation Techniques

- ⇒ Animation techniques are:
 - Animation Process
 - Cell animation
 - Computer animation

Steps to be followed in creating animation are:-

- Organize the animation in a series of logical steps.

- o Choose an animation tool best suited for the job.
- o Build and tweak the sequences
- o Post-process the completed animation

Cell Animation

- o Cell animation is the traditional form of animation.
- o Cell animation is the art of creating 2D animation by hand on sheets of transparent plastic called "cels".
- o The animation techniques a made famous by Disney
- o Cell animation is a technique in which a series of progressively different graphics are used on each frame of movie film.
- o Cell animation begins with keyframes. The term "cel" is derived from the clear celluloid sheets that were used for drawing each frame.
- o Also called hand-drawn animation

- o Cell animation begins with keyframes.

Keyframes:-

- o Keyframes refer to the first and the last frame of an action.
- o The series of frames in between the keyframes are drawn in the tweening process.
- o A keyframe defines the starting and ending point of any smooth transition.

Tweening:-

- o Tweening is an action that requires calculating the number of frames between keyframes.
- o It is also known as "inbetweening"
- o It is the process of creating the inbetweens, which are the images that go between keyframes.
- o It depicts different points in an action.
- o Tweening is necessary to convey a sense of fluid movement with still images.

Interpreting Characteristics:

- The drawing between keyframes
- Used to convey smooth motion.

Pencil test:

- A pencil test is the process where an animator digitally captures rough, pencil drawings to check how their animation is working.

- In short, 'cel animation' is

generally observed as the act of creating 2D animation by human on transparent sheets called cel.

Computer animation

- Computer animation is electronically generated movement of anything on computer screen.
- Computer animation is same like cel animation.
- But there is primary difference between cel animation and

Computer animation.
⇒ Computer animation is also called digital animation.

Levels of computer animation:

⇒ There are different levels of digital animation are:

- Basic
- Intermediate
- Advanced

Inks:

- it is special method for computing color values, providing edge detection, and layering so that images can blend or like wire mix their colors to produce special transparencies, inversions & effects.

Kinematics:-

Kinematics is the study of the movements and motion of structures that have joints, such as a walking man.

it is the process of linking objects and defining their relationship and limits.

Inverse kinematics is available in high-end 3-D programs such as Lightwave and Maya.

Morphing

Morphing is a popular effect in which one image transforms into another.

it is the most complicated transformation.

Morphing is commonly used in movies, cartoons, advertisements and computer games.

The term morphing comes from the word "metamorphosis".

it is the process of using motion pictures and adding some animations to already existed image. we can create a new image.

The morphed images built at a rate of eight frames per second with each

Cell Animation Vs Computer Animation

Cell animation	Computer animation
- It is created by hand onto cell-which sheets, known as animated cels. - it use physical materials & activities.	- Digital animation is created by software & computers. - uses virtual materials in digital space.

Warping

it is an effect allows you to manipulate a single image.

warping means that points are mapped to points without changing the colors.

warping assigns a new position in the target image to each point of the original image.

Twisting transfers each point in the original image to the new position calculated by warping.

Date: _____

Animation Process:

Animation is a systematic activity with a step-by-step procedure. Steps that are followed in Animation process are:

(1) Research & brainstorming:

- The creators, writers and animators come up with an idea for the story of the animation.
- Research is done to look - the specific aspects of the plot. The creator thinks:
 - what will be the subject of the animation.
 - who will be the primary & secondary characters
 - when and where the story will be set.
- They chalk out where the story is going to begin and it would end.

2) Scripting & Storyboarding:

- A sequential outline of the story is written with dialogues for each character and the happenings and occasions that are going to take place in every scene so a script is essential. A technical form of how the animation is going to look like.
- Based on the script, a story board is made or computer aided 2D sketch of every scene is created known as the storyboard. The storyboard is same as the basis of all further animator steps.

3) Dialogue recording:

- To give life to the characters, voice over artists record the dialogues of different characters reading from the script. They add to the personality, expression and specific accent of the characters and this helps the animators to create exact mouth movements of the characters as they speak.

4) Animation Production:

- At this stage, the actual job of animation begins. Layout of the animation is created showing the character size, shape & environment.
- 1) Geometric surface representation of each object is created in modeling.
 - 2) Texturing is done to add surface properties to various objects.
 - 3) Rigging involves putting a bone structure into an object so that animator can move specific parts of that object.
 - 4) Lighting is done to set a certain mood for a scene or shadows of the objects & characters.
 - 5) Finally, all elements of the animation such as characters, objects, backgrounds etc are separated into different layers in the process of rendering.

Date: _____

5) Animation Post Production:

In this final stage, Compositing is performed which puts together the different layers rendered previously.

Visual effects such as dust, raindrops, sparks, blasts etc are added into the animation if required.

Sound effects:

- for different objects
- Events of the animation

are recorded.

In the end, the animation is combined with the sound and the VFX and final touches are given in video editing.

Ultimately, the video is converted into the required video format, ready for use in different media.

What is path animation?

Path animation involves moving an object on a screen that has a constant background.

e.g. a cartoon character may move across the screen regardless of any change in the background or the character.

- it does not allow to change the shape of objects while it is moving.
- it moves an object along a predetermined path on the screen.
- The path can be a straight line or have a number of curves.
- + keyframes & Tweening.
- Also known as vector based animation.